

he supported the idea of eugenics, and so on. However, personal follies must bow before intellectual greatness: he was awarded the Nobel Prize for physics in 1956.

Intel today produces billions upon billions of transistors every day, but those three earned very little. One-hundredth of a paisa per transistor in royalties would have meant...

1948

The SSEM (Small Scale Experimental Machine) was built at Manchester University, based on ideas from John von Neumann, a Hungarian Mathematician, about stored-program computers. All modern computers use the von Neumann architecture, as it is called; find out more in chapter 2. Essentially, this was the first computer ever to store both programs and data in memory!



Shockley (seated), and Bardeen and Brattain, in 1947

1949

Another first, but a *relatively* minor one: Maurice Wilkes and his team at Cambridge University build a computer called the EDSAC. It used paper tape for input and output, but it was the first stored-program computer that operated a regular computing service.

Proposed in 1949 by John von Neumann and completed in 1952 at the Institute for Advanced Study, Princeton, the EDVAC (Electronic Discrete Variable Computer) was the first computer to use magnetic tape. A breakthrough of sorts, because earlier computers could only be reprogrammed by re-wiring them! The EDVAC gave users the luxury of loading new programs off the tape.